UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00716

LH OBSERVATIONAL COSMOLOGY

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. Consider our Universe:

   (a) State its current composition, giving approximate values for the different components. **[4]**

   (b) Give an expression for the Hubble distance d_H at the present epoch and estimate its numerical value. Show that galaxies separated by a proper distance larger than the Hubble distance will have a recession speed $v_H > c$. **[4]**

   (c) What is the physical meaning of the critical density ρ_c? Starting from the Friedmann equation as given in the formula sheet, derive an expression for ρ_c. Compute its present-day value and discuss its implications for the Universe. **[6]**

   (d) Calculate the redshift $z_{m\Lambda}$ at which the matter-Λ equality occurred. How much time has passed between $z_{m\Lambda}$ and today?
   The following equation will be useful, where all symbols have their usual meaning.

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m,0}}} \ln\left[\left(\frac{a}{a_{m\Lambda}}\right)^{3/2} + \sqrt{1+\left(\frac{a}{a_{m\Lambda}}\right)^{3}}\right]. \qquad (1)$$

   [10]

   (e) Derive an explicit expression for the scale factor $a(t)$ for the future evolution of our Universe. Clearly state any assumptions you make. **[6]**

2. The cosmic microwave background and structure formation in our Universe:

   (a) Determine the temperature of CMB photons at the time of matter-radiation equality. **[4]**

   (b) Sketch the angular CMB power spectrum and qualitatively explain its main features. **[4]**

   (c) The smallest temperature fluctuations resolved by the *Planck* satellite have an angular size of $\delta\theta \approx 5$ arcmin. Calculate their physical size at the surface of last scattering in kiloparsecs. [Hint: Make the simplifying assumption that the Universe has always been matter-dominated. An arc minute is $0.000290888\,\mathrm{rad}$.] **[6]**

   (d) Explain in detail why the isotropy of the CMB is inconsistent with the standard Big Bang model and how inflation can solve this problem. **[4]**

   (e) Describe qualitatively how the CMB power spectrum is linked to the galaxy correlation function. What is the BAO "bump"? **[2]**

   (f) The evolution of density fluctuations δ is governed by the following equation:

   $$\ddot{\delta}(t) + 2\dot{\delta}(t)H(t) - \frac{3}{2}\Omega_m(t)H^2(t)\delta(t) = 0,$$

   where Ω_m and H have their usual meaning.
   Discuss briefly the role of the second term.
   Derive the growing mode solution assuming matter-domination and spatial flatness. **[10]**

3. (a) Consider an expanding universe:

(i) Describe the cosmic distance ladder. Explain two rungs of the ladder in detail. **[5]**

(ii) Show that the current proper distance to a galaxy with redshift z is given by:

$$d_p(t_0) \approx \frac{c}{H_0} z \left[1 - \frac{1+q_0}{2} z \right],$$

where the variables in the equation have their usual meanings.
Discuss the implications for Hubble's law and compare to observations.
The following equation will be useful:

$$\frac{1}{a(t)} \approx 1 - H_0(t-t_0) + \left(1 + \frac{q_0}{2}\right) H_0^2 (t-t_0)^2.$$

[10]

(iii) Show that for a flat universe $d_L = d_p(t_0)(1+z)$. **[5]**

(b) Dark matter is believed to be an important component of the Universe.

(i) List the evidence for dark matter in the Universe. Discuss two observations in detail. **[8]**

(ii) Can neutrinos account for dark matter? Justify your answer. **[2]**

Formula Sheet

Friedmann-Lemaître-Roberston-Walker (FLRW) metric:

$$ds^2 = -c^2 dt^2 + a^2(t)\Big[dr^2 + S_\kappa(r)^2(d\theta^2 + \sin^2\theta d\phi^2)\Big]$$

where

$$S_\kappa(r) = \begin{cases} R_0 \sin(r/R_0) & \kappa = +1 \\ r & \kappa = 0 \\ R_0 \sinh(r/R_0 & \kappa = -1 \end{cases}$$

Friedmann equation (general case):

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H^2(t) = \frac{8\pi G}{3}\Big(\rho_r(t) + \rho_m(t)\Big) - \frac{\kappa c^2}{R_0^2 a^2(t)} + \frac{\Lambda}{3}$$

Acceleration equation (general case):

$$\frac{\ddot{a}(t)}{a(t)} = -\frac{4\pi G}{3}\sum_{i\in\{r,m\}}\left(\rho_i(t) + \frac{3p_i}{c^2}\right) + \frac{\Lambda}{3}$$

Fluid equation (general form) for the i-th density component:

$$\dot{\rho}_i(t) + 3\frac{\dot{a}(t)}{a(t)}(1 + w_i)\rho_i(t) = 0$$

Deceleration parameter for N density components:

$$q_0 \equiv -\left(\frac{\ddot{a}}{aH^2}\right)\bigg|_{t=t_0} = \frac{1}{2}\sum_{i=1}^{N}\Omega_{i,0}(1 + 3w_i)$$

Planck energy spectrum:

$$E(\nu, T) = \frac{h\nu}{e^{(h\nu)/(k_B T)} - 1}$$

Energy density of a thermal gas:

$$\rho(T) = \frac{8\pi^5 k_B^4}{15 h^3 c^3} T^4$$

CMB power spectrum:

$$C(\theta) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta) \quad \text{and} \quad C_\ell = 2\pi \int_{-1}^{1} C(\theta) P_\ell(\cos\theta)\, d\cos\theta$$

Jeans length:

$$\lambda_J = 2\pi c_s t_{\rm dyn} = 2\pi \left(\frac{2}{3}\right)^{1/2} \frac{c_s}{H} = 2\pi \left(\frac{2}{3}\right)^{1/2} \sqrt{w} \frac{c}{H}$$

Relativistic equation governing the growth of small matter perturbations δ:

$$\ddot{\delta}(t) + 2H(t)\dot{\delta}(t) - \frac{3}{2}\Omega_m(t) H^2(t) \delta(t) = 0$$

Saha equation:

$$\frac{1-X}{X^2} = 3.84\eta \left(\frac{k_B T}{m_e c^2}\right)^{3/2} \exp\left(\frac{Q}{k_B T}\right)$$

Neutron-to-proton ratio:

$$\frac{n_n}{n_p} = \exp\left(-\frac{(m_n - m_p)c^2}{k_B T}\right)$$

Klein-Gordon equation:

$$\ddot{\phi}(t) + 3H(t)\dot{\phi}(t) = -\hbar c^3 \frac{dV(\phi)}{d\phi}$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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